

MU-HUA (MAURICE) WANG

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RESEARCH INTEREST

Physical assistive robotics for long-term care. I want to build robots that carry out contact-rich care tasks on a bedridden human body, starting with diaper changing and the cleaning that goes with it. The task demands three things at once: perceiving a body that will not cooperate, knowing enough about that particular person to move them without harm, and handling deformable material safely against skin. My work so far covers the first two and has begun the third. I built and clinically validated a markerless human motion capture system. I have studied, in musculoskeletal simulation, what a robot must know about a person before it moves their limb and what a learned model of a body is worth to a learner. I am now building a simulation platform for robot-assisted diaper changing. I am looking for a lab where these can be assembled into a machine that does the task.

EDUCATION

National Yang Ming Chiao Tung University (NYCU) — Taipei, Taiwan

M.S., Biomedical Engineering — entered Feb 2025; on approved leave of absence since May 2025 (enrollment retained)

Took leave to move from measuring human movement to building systems that act on it. Taught full time through the leave while starting the robot-learning work below, and now research full time.

B.S., Biomedical Engineering — Sep 2020 – Jun 2024

- GPA 3.72 / 4.30 · Final two years 4.07 / 4.30
- **IELTS 8.0 overall** — Reading 9.0, Listening 8.5, Speaking 7.5, Writing 7.0
- Coursework: Deep Learning for Biomedical Data Analysis; Neural Engineering; Brain and Physiological Signal Analysis; Smart Medical Device Mechanical Design; Medical Device Regulatory Affairs

RESEARCH EXPERIENCE

Project pages, videos and full manuscripts for everything below: maurice1128.github.io

I. Care-Robot Simulation Platform — Independent

2026 – present · in progress

Measuring Whether a Care Robot Hurts the Patient: a simulation platform for robot-assisted diaper changing and perineal cleaning

Existing caregiving simulators ask whether the robot can finish the task. This platform asks what the robot does to the body while trying.

- **Three axes, scored on every run:** is the garment on, is the perineum cleaned, is the patient injured.
- **The patient.** An SMPL-X body driven by MuJoCo, with pelvis, sacrum and femur meshes registered under the skin (hip-centre residual 10.4 mm). Every care position (supine, side-lying, frog-leg) is settled physically and must pass gates before it is used.
- **The tissue model.** A soft-tissue model of the sacral region with bone on the deep side. Its solver reproduces closed-form compression tests to 1.8×10^{-5} and its strain fields are converged across three mesh densities.
- **Tissue load.** After 285 simulated minutes supine, changing the mattress from rigid to 30 kPa foam lowers skin pressure by about 15% but shrinks the volume of deep tissue above the 285-minute strain level from 69 to 4 cm³. Interface pressure alone does not tell you about deep tissue.
- **Cleaning.** Of 798 legal robot bases, 19 reach all four targets, with peak contact force at most 2.8 N and no skin penetration. With a compliant cloth pad at 13–15 mmHg, 0.93 of the main perineal region is wiped at least three times under a criterion fixed before the run.
- **Garment.** A brief is closed around the settled body: front panel pulled up with stretch 1.24×, all four side tabs fastened.
- **Scope.** Simulation only, on a statistical body model; results are model-internal and not validated against measurements on people. Ongoing. Video and figures on the [project page](#).

II. Robot Learning and Embodied AI — Independent Research

Oct 2025 – present · Sole author on five manuscripts, two submitted

A self-directed programme, without an advisor or a lab, on one question: what must a robot know about a body, its own or the one it is touching, and what does that knowledge buy? Study design, experiments, statistics and verification are my own. Each result is tested against controls built to break it, and what it did not show is reported as a finding.

Gentle by knowing: person knowledge, not force thresholds, protects restricted joints in simulated robotic arm lifting — *submitted, Frontiers in Robotics and AI (Brief Research Report)*

- Robot safety standards and recent limb-moving robots rely on contact-force limits, but when a robot supports a multi-joint limb from outside, the loaded joint is not on the axis of its force sensor. I made the robot's knowledge of the person the variable: in a MyoSuite musculoskeletal simulation a capsule supports the forearm from below and raises it 5 cm for simulated persons with restricted shoulders or elbows; controllers differ only in what they know and whether they may decline. A harmful outcome is a robot-caused end-range joint load of at least 0.5 N·m, a mechanical-exposure proxy.
- No force-magnitude guard reduced harmful outcomes by significantly more than the 5 percentage points fixed in advance, at four thresholds, under a hard limit or graded end-range resistance. Guards mostly never trip before the joint is loaded, and when they trip the load has already reached 0.5 N·m; 95–100% of harmful outcomes occur during the first attempt; every harmful fixed-rule lift stays under the ISO/TS 15066 lower-arm force limit of 160 N.
- With a complete record carrying 20° error, a policy learned offline in simulation reduced harmful outcomes from 110 to 25 of 300 new simulated persons by declining 91 of them, with 174 vs 177 safe completions. The reduction replicated on further persons (28 vs 108 harmful of 300) and held under posture noise, over-estimated records, a shifted population (38 vs 170 of 250), graded resistance and two physics perturbations. Knowledge helped mainly by telling the robot whom not to lift; post hoc, knowing which joint is restricted plus the settled posture suffices.
- Reported against my own result: everything is simulated and harm has no dose–response; safe-completion non-inferiority to the fixed rule was not established (-0.01 [-0.05 , $+0.03$]), and at a 2.5 N·m threshold the policy completes 6–8 points fewer lifts; with half the restrictions unrecorded the reduction shrinks from -0.27 to -0.15 and unrecorded persons are harmed at the fixed-rule rate; against perfect knowledge (237 safe, 0 harmful) 47 of the 63 safe completions it missed were unnecessary refusals and 18 of its 25 harmful outcomes were persons it should have declined; the protocol was kept locally, not externally registered.

Withdrawn Teachers and Learned Body Models in Muscle-Driven Motor Learning — *submitted, IEEE TCDS*

- Two questions on one soft actor-critic learner and six simulated muscle-driven bodies (a MyoSuite elbow and finger; planar arms of one to four joints driven by 2, 4, 8 and 10 MuJoCo muscles): when a teacher is withdrawn, is the learner left behind one never guided, and is the drop lost knowledge or the changed objective? When a babble-learned body model helps, is it because of what it learned? Twelve matched seeds per condition, twenty-four on the three- and four-joint arms; 1,056 runs.
- Controls that change one thing at a time: at withdrawal the teacher is replaced by a frozen copy of the student's own actor, which keeps the imitation term and removes only the teacher; a forward model trained once on reward-free babbling is compared with a randomly initialised model in identical Dyna machinery.
- Finger, competent teacher: the withdrawn learner ends 43 mm ahead of one never guided (-43.25 [-54.72 , -31.79] mm pooled over seven withdrawal times, against a teacher worth -61.71 mm) and keeps more the longer it was taught: 24% of the teacher's worth when withdrawn at 2,000 steps, 101% at 8,000. From 4,000 steps of attachment on, 65–94% of the drop after withdrawal is the removed imitation term, not the teacher's unabsorbed knowledge. Elbow: the teacher is worth 8.65 mrad and its benefit goes with it; no deficit against never being guided is detected at 100k or 200k steps.
- Planar arms: a random model is not harmful on one joint and increasingly harmful after (+22, +97, +262 mm against model-free learning on two, three and four joints); the babble-trained model avoids this harm, and beats the random one on five of six bodies under both endpoints. All ten primary tests hold after Holm correction (largest corrected $p = 0.0018$).
- Reported against my own result: single-task reach in simulation on deterministic, fully observed bodies, with the imagined reward from exact forward kinematics; the finger teacher is the best of twelve candidate runs and one student shares its seed; the replay buffer keeps guided-phase transitions, so part of what the withdrawn learner keeps is buffered guidance; the teacher component of the drop shrinks partly by construction; the endpoint, the retention threshold and the ten-test family were fixed during revision, after data had been seen; the elbow comparison is not at a plateau.

Learned In-Betweeners Make Simulated Dexterous Hands Learn Faster — *manuscript, prepared for IEEE RA-L*

- Modular motion models such as MotionBricks build motion from reusable parts, but prior work on synergies, latent action spaces and skill chaining changes the prior, the decision rate, the data and the training procedure

together. I tested each of the model's two layers on simulated hands, with the main comparisons declared before the runs, to find where a modular gain comes from.

- Motion level: each motion prior became the action space of a PPO policy on a simulated Shadow Hand holding a cube under a 1.5 g lateral acceleration, every prior deciding once per 16 frames. Keyframes joined by a learned in-between score 0.59 against a continuous latent's 0.22 on skill sequences never seen in training at 1.5 M frames (20 seeds, 17 of 20; difference +0.38, 95% CI [0.23, 0.52]) and reach their training plateau sooner. At 5 M frames the data do not separate the two (0.44 against 0.49, 10 seeds).
- The gain belongs to the learned in-between, not to modularity as such: the same keyframes joined by straight lines never learn to release (0 of 10 seeds), and a bank of 16 fixed primitives is the weakest prior (0.08).
- Skill level, on Adroit relocate extended to grasp-then-carry with behaviour cloning from 25 demonstrations: selecting the best policy for each stage completes 97 of 400 fresh test episodes against 61 for the best single policy ($p = 0.002$); separately trained skills do not help (45 of 400, $p = 0.12$) and lose to per-stage selection ($p < 10^{-5}$).
- Reported against my own result: everything is in simulation, one task per layer; the declared prediction that separately trained skills beat the best flat policy failed; the early lead is significant on the cube only (three other objects, $p = 0.067$); the comparison with plain PPO changes the decision rate as well as the prior; the skill-level pipeline runs once, so the pool analysis (per-stage selection beats separate training in 99% of 184,756 matched pools) shows the result does not hinge on pool size, not that it replicates. The four conditions under which separately trained modules would pay are offered as hypotheses; neither task meets the fourth.

Partial labels hide compositional difficulty in hand-motion data — *manuscript, prepared for TMLR*

- Held-out label combinations are the standard test of whether a motion model generalises to unseen compositions. On recorded motion the split is made on labels, and the prior work using this split on recorded hand data had not checked whether holding out a label also holds out the motion.
- Measured a paired penalty, the share of reconstruction error a prior removes by seeing the held-out combinations, on four public hand-motion datasets: OakInk-Image, GRAB, OakInk2 and TACO. Calibrated it against a synthetic zero-truth control (+2.6% of the naive error), a planted interaction recovered on synthetic data (+9.1%) and on GRAB (+13.2%), and permuted label grids on three of the datasets (+3.7% OakInk-Image, +1.4% GRAB, -3.4% TACO). The floor of the measurement depends on the dataset.
- On OakInk-Image, joining clips so that each correct label described only the first half of a trajectory, with the number of trajectories, the categories and the training volume matched, reduced the penalty from +17.0% to +3.9%, at most 2.9 points above the control (difference +0.9, 95% CI -1.1 to 2.9). The windows the label did describe lost 82% of their penalty above the control (95% CI 65% to 97%), so the change lies mostly in the training sets.
- A penalty on TACO, written down before its data were opened, was found: +10.6%, 8.0 points above the control.
- Reported against my own result: per-seed estimates of the TACO penalty ranged from -3.8% to +27.4%; the error is an autoencoding error, so the result is about passing an unseen composition through a bottleneck, not about generating one; one prior architecture was used, and the manipulation was run on one dataset; only one size of planted interaction was run; the written-down settings are internal documents with file dates, not registrations time-stamped by a third party.

Reparameterisation Invariance Limits the Benefit of Reorganising a Body Code for Morphology Transfer — *draft*

- An algebraic identity rules out a whole family of "reorganise the code" proposals before any experiment runs; the two conditions it needs are then measured. A re-encoding that preserves decodability by construction still costs +0.875 mm, a counterexample to decodability as a proxy for representation quality.

III. RGB Multi-View 3D Markerless Motion Capture — NYCU

Research Assistant, Oct 2023 – Sep 2024 · Team Lead, Oct 2024 – Sep 2025 Advisor: Prof. Hui-Ting Shih

Real-time whole-body pose sensing for clinical use — the perception layer a robot working over a bed has to have. I wrote the pipeline, ran the validation experiments, and collected the clinical data myself.

- **Built the real-time pipeline end to end.** GPU acceleration and CPU parallelism across capture, inference, triangulation and inverse kinematics; DLT triangulation with occlusion handling; and **ROWV**, an outlier-rejection algorithm for small-sample settings, faster and more accurate than KNN and RANSAC on this problem.
- **Ran the validation experiments myself.** VICON comparison over 21 participants, plus a study over 104 walking trials from 11 participants on a nine-camera ring showing that per-joint verdicts between triangulation variants depend on the joint-centre offset they contain.
- **Ran the field data collection.** 200+ stroke and knee-pain patients (Dr. Ho, Vietnam); 40+ stroke patients with exoskeleton (Kaohsiung Chang Gung Memorial Hospital); 20+ elderly participants (Kaohsiung daycare centre); real-time feedback trials with lower-limb amputees (NYCU Physical Therapy and Assistive Technology). Hospital wards, clinics and daycare centres, not a laboratory.

- **Asked how much sensing error the downstream decision can absorb** — forward-dynamic simulation with a Geyer–Herr reflex controller over 66 design cells, showing identification of a weak muscle group falls from 0.845 to 0.549 as joint-angle error grows $0^\circ \rightarrow 12^\circ$.
- **Shipped the whole product**, not only the algorithm: clinician-facing GUI (PyQt6), SQL and cloud pipeline with ICD-10 coding for longitudinal and cross-sectional search, and an Unreal Engine rehabilitation game.
- **Led a 5-person R&D team** in the second year and drove the system into clinical deployment: 4 conference presentations and 6 media features (CNA, PTS News, Yahoo! News Taiwan, iThome, OCAC, NYCU).
- **Ran the collaborations**, across institutions that do not normally open up to a student: 2 signed MOUs (Taipei Veterans General Hospital; Taipei City Hospital), Kaohsiung Chang Gung Memorial Hospital, the Institute for Information Industry (cloud patient database with ICD-10 coding), an international site in Vietnam, a community daycare centre and a private clinic in Kaohsiung, a sports-performance company, and NYCU Physical Therapy and Assistive Technology.

IV. Circular-Geometry Representation of Behavioural Trajectories — Taoyuan General Hospital

Jan 2023 – present · Advisor: Dr. Tso-Jeng Wang

A methods problem disguised as a clinical one: the discriminating feature of a dose trajectory is its *shape*, but shape is confounded with magnitude and lives on a space that Euclidean clustering gets wrong.

- **Built the representation.** Normalised each daily consumed-dose series to its own value seven days earlier, took the extreme three-day moving-window slopes on each side of the test, and mapped the (post, pre) slope pair to a polar angle $\theta = \text{atan2}(\text{post}, \text{pre})$. Shape becomes comparable independent of absolute magnitude, and the feature space becomes a torus.
- **Clustered on the manifold, not in the plane.** Fitted a bivariate (product) von Mises mixture by EM on the circle, because Euclidean clustering does not respect wrap-around at $0^\circ/360^\circ$. For the stable-dose branch, reduced 114 variables by PCA and cut a circular kernel-density estimate at its valleys. Nine patterns emerged, spanning positive/negative occurrence ratios 1.64 to 0.50.
- **Attacked my own result.** Outcome-blind re-clustering, 500 bootstrap refits per branch, within-patient conditional logistic regression, restriction to windows containing no interpolated day, and reporting across $K = 4-8$ because K was an investigator choice and no criterion selected it.
- **Reported what did not survive.** Post-test escalation magnitude (OR 1.36 $\rightarrow$ 1.06 within patients) is not claimed; the stable-dose branch is labelled exploratory; the association did not replicate across a temporal split; and the trajectory adds essentially nothing predictive over attendance. Pre-test escalation did survive, and strengthened (1.15 $\rightarrow$ 1.26 per SD, $p = 3 \times 10^{-7}$).
- Scale: 9,900 analysable urine tests from 552 patients over 2018–2022, from a 20,108-record extract; earlier phase covered 1,000+ patients from the Taiwan National Health Insurance Database.

V. EEG and EMG Sleep–Epilepsy Semi-Automated Analysis — NYCU

Oct 2022 – Oct 2023 · Advisor: Prof. You-Yin Chen

- Built a signal pipeline processing 24-hour recordings in 3 minutes (detrending, downsampling, filter design, time–frequency conversion, outlier detection) with automatic WAKE / NREM / REM staging from band-specific energy features, plus a manual-scoring GUI for cross-labeler consistency.
- Designed a surgical modification that raised EMG signal-to-noise ratio at the source. Collaboration with Hualien Tzu Chi Hospital and Taipei Medical University.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed conference papers

1. **Wang, M.-H.**, Tsai, K.-T., Chen, B.-H., Huang, L.-W., Tsai, C.-H., Hsu, C.-S., & Chen, Y.-Y. *Markerless Gait Analysis for Lower Back Pain: A Cost-Effective Solution to Assess Pelvis–Trunk Coordination*. **SEMBA 2025**. (first author)
2. Huang, L.-W., **Wang, M.-H.**, Chen, B.-H., Su, K.-H., Tsai, C.-H., Chen, Y.-Y., & Shih, H.-T. *A Real-Time Algorithm for Detecting Gait Events in Lower-Limb Prosthesis Users Using a Markerless Motion Capture System*. **ASB 2025**.
3. Huang, A.-Y., **Wang, M.-H.**, Huang, L.-W., Chen, B.-H., Wang, H.-H., Kuo, C.-H., Lu, S.-W., & Lo, Y.-C. *Spatiotemporal Cortical-Synergy Connectivity in the Gait Cycle*. **SEMBA 2025**.
4. Chang, C.-W., **Wang, M.-H.**, Lin, Y.-C., Wang, C.-Y., Li, S.-J., Chen, T.-C., Liang, Y.-W., Chen, C.-T., Chen, Y.-Y., & Lin, S.-H. *Investigating the Effect of Central Thalamic Deep Brain Stimulation on Sleep in an Alzheimer's Disease Model*. **ISMRM 2024**.

5. Chen, B.-H., Tsai, C.-H., Huang, L.-W., **Wang, M.-H.**, Hsu, C.-S., & Chen, Y.-Y. *Robust Self-Occlusion Mitigation for Enhanced Accuracy in Multi-View 3D Markerless Motion Capture Systems*. **ICMST 2024**.

Submitted and prepared manuscripts

- S1. **Wang, M.-H.** (sole author). *Gentle by knowing: person knowledge, not force thresholds, protects restricted joints in simulated robotic arm lifting*. **Frontiers in Robotics and AI** (Brief Research Report), submitted 2026.
- S2. **Wang, M.-H.** (sole author). *Withdrawn Teachers and Learned Body Models in Muscle-Driven Motor Learning*. **IEEE Transactions on Cognitive and Developmental Systems**, submitted 2026. Merges and supersedes the earlier draft *Two Routes to a Motor Prior*.
- S3. **Wang, M.-H.** (sole author). *Learned In-Betweeners Make Simulated Dexterous Hands Learn Faster*. Manuscript prepared for **IEEE Robotics and Automation Letters**, 2026.
- S4. **Wang, M.-H.** (sole author). *Partial labels hide compositional difficulty in hand-motion data*. Manuscript prepared for **Transactions on Machine Learning Research**, 2026. Supersedes the earlier draft *Which Hand-Motion Composition Axes Are Worth Measuring*.

Manuscript drafts — on the website

Robot learning

- D1. **Wang, M.-H.** (sole author). *Reparameterisation Invariance Limits the Benefit of Reorganising a Body Code for Morphology Transfer*.

Human motion sensing and clinical biomechanics

- D2. **Wang, M.-H.**, & Shih, H.-T. *Identification of a Weak Muscle Group Tolerates More Joint-Angle Error than Kinematic Reliability Conventions Allow*.
- D3. **Wang, M.-H.**, & Shih, H.-T. *Joint-Centre Offset Dominates 3D Position Error and Can Reverse Comparisons Between Markerless Triangulation Methods*.
- D4. **Wang, M.-H.**, & Shih, H.-T. *Distinguishing Graded Plantarflexor Hyperreflexia from Graded Dorsiflexor Weakness in Simulated Post-Stroke Gait: A Search over 81 Sagittal Kinematic Variables*.

Behavioural data science

- D5. **Wang, M.-H.**, & Wang, T.-J. *Consumed Methadone Dose Patterns Around Randomly Scheduled Urine Testing: Pre-Test Escalation and Higher-Risk Trajectories in Long-Retained Maintenance Patients*.

TEACHING

Physical Science Teacher (Grade 8) — Tamkang High School, New Taipei · Feb 2026 – Aug 2026

Taught the full Grade-8 physical science curriculum to two of the grade's eight classes.

- **Class 8 finished 2nd of eight classes on the third term exam, up from 3rd**, with its mean margin over the school average widening from +5.1 to +11.8 points across three term exams.
- **The gain came from the bottom of the distribution, not the top.** By the third exam, Class 8's bottom-quartile cutoff (78) had reached the school-wide median, and its bottom-12% cutoff stood 28 points above the school's — up from a 14-point gap on the first exam.
- Class 8 held the second-narrowest score spread in the grade (SD 15.07 against a school-wide 21.86): the class improved by pulling up students who were behind, not by stretching students already ahead.
- **Designed the mechanism that produced it.** The class earned its reward only if *every* student scored 100 — a 10-minute paper, then open discussion, then a fresh paper, repeated until the whole class cleared it. The collective condition put the strongest students to work teaching the weakest, which is where the measured gain came from.
- **Ran self-initiated weekly remedial sessions** outside class hours across three grades, targeted at the students who were behind — not assigned by the school.
- Ran hands-on chemistry and physics activities outside the syllabus, and class events to build cohesion.

HONORS AND AWARDS

2nd Place — Youth AI Startup Collaborative Hackathon, Ministry of Digital Affairs · Fall 2023 · Taipei

Led a team building the business model and commercialization path for a markerless visible-light motion capture system. Prize US\$1,000.

1st Place — Competition for Innovative Proposals in Elderly Leisure, Health and Technology Practices, National Chi Nan University · Fall 2023 · Nantou

Proposed a fall-detection and emotional-gait-recognition model for elderly care. Prize US\$340.

TECHNICAL SKILLS

Languages	Python (advanced), MATLAB (advanced), C/C++ (intermediate)
ML / RL	PyTorch, TensorFlow; SAC, PPO and Dyna-style model-based RL; policy distillation; offline policy learning; world models; behaviour cloning; CNN/VGG; LSTM
Robotics & simulation	MuJoCo; MyoSuite; Shadow Hand; Adroit; SMPL-X; musculoskeletal forward dynamics; Geyer–Herr reflex control; inverse kinematics; DLT triangulation; Unreal Engine
High-performance computing	CuPy, Numba, multiprocessing, GPU acceleration, CPU affinity and priority tuning
Statistics & experiment design	matched-seed paired designs; predictions fixed before the runs; zero-truth, planted and permuted-label controls; adversarial controls; non-inferiority and equivalence margins; von Mises / directional statistics; bootstrap reproducibility; conditional logistic regression
Systems	PyQt6, SQL, cloud data pipelines, ICD-10 clinical coding

LEADERSHIP AND ACTIVITIES

- **Leader, Rural Service Team** · Fall 2024 – Fall 2025 — led two 3-day service camps in Tamsui and Fenglin combining educational workshops with community service. **Programme lead, Beitou church service team** (2026).
- **Music Director, church worship team** · present — and **Founder & President, ToGather Music (YouTube)** · Fall 2021 — a team of 3 producers and 5 vocalists, a self-built recording studio, 13 original songs published. Five years of continuous responsibility for a performing team.
- **Long-term mentor** to a group of four students spanning primary to senior-high school.
- **Chief Organizer, Biomedical Engineering & Nursing Student Camp** · Fall 2022 — 50 students across two disciplines. **Graduation Representative** · Fall 2024.